AI-Driven Development Life Cycle: Reimagining Software Engineering

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Source: https://aws.amazon.com/blogs/devops/ai-driven-development-life-cycle/
Setup: https://github.com/awslabs/aidlc-workflows
Adaptation: Ricardo de Luna Galdino, EngSoft
Download: ai-dlc_aws-article_raja-sp.pdf

Business and technology leaders are constantly striving to improve productivity, increase velocity, foster experimentation, reduce time-to-market (TTM), and enhance the developer experience. These North Star goals drive innovation in software development practices. This innovation is increasingly being powered by artificial intelligence. Particularly, generative AI powered tools such as Amazon Q Developer and Kiro have already begun to revolutionize how software is created. As things stand, organizations employ AI in software development through two primary approaches: AI-assisted development, where AI enhances specific tasks like documentation, code completion, and testing; and AI-autonomous development, where AI is expected to generate entire applications without human intervention based on user requirements. Both these approaches have produced suboptimal results in terms of velocity and software quality that AI-DLC aims to address.

Why do we need a transformative approach to AI in software?

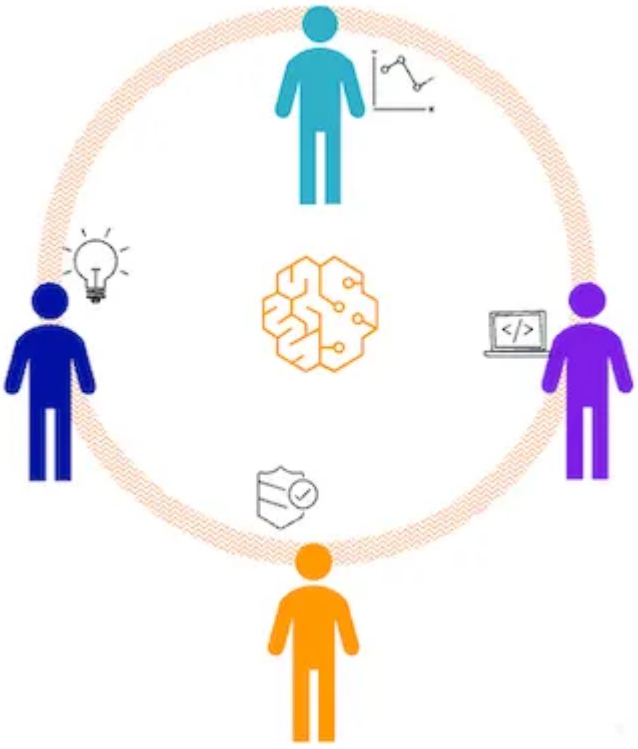
Our existing software development methods, are designed for human-driven, long running processes, with product owners, developers, architects alike spending most of their time on non-core activities such as planning, meetings, and other software development lifecycle (SDLC) rituals. Simply retrofitting AI as an assistant not only constrains its capabilities but also reinforces outdated inefficiencies. To truly harness AI's power and achieve the productivity North Star goals, we need to reimagine our entire approach to the software development lifecycle.

To achieve transformative results, we need to position AI as a central collaborator and teammate in the development process, and leverage its capabilities throughout the software development lifecycle. This is why we're introducing the AI-Driven Development Lifecycle (AI-DLC), a new methodology designed to fully ingrain AI capabilities into the very fabric of software development.

What is AI Driven Development Life Cycle (AI-DLC)?

AI-DLC is an AI-centric transformative approach to software development that emphasizes two powerful dimensions:

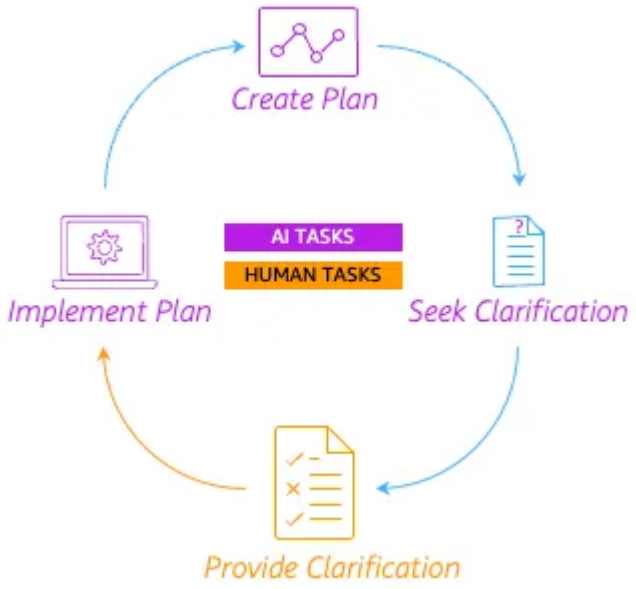
- **AI Powered Execution with Human Oversight**: AI systematically creates detailed work plans, actively seeks clarification and guidance, and defers critical decisions to humans. This is critical since only humans possess the contextual understanding and knowledge of business requirements needed to make informed choices.
- **Dynamic Team Collaboration**: As AI handles the routine tasks, teams unite in collaborative spaces for real-time problem solving, creative thinking and rapid-decision-making. This shift from isolated work to high-energy teamwork accelerates innovation and delivery.



These two dimensions enable you to deliver software faster without compromising on quality.

How does AI-DLC work?

At its core, AI-DLC operates by having AI initiate and direct workflows through a new mental model:

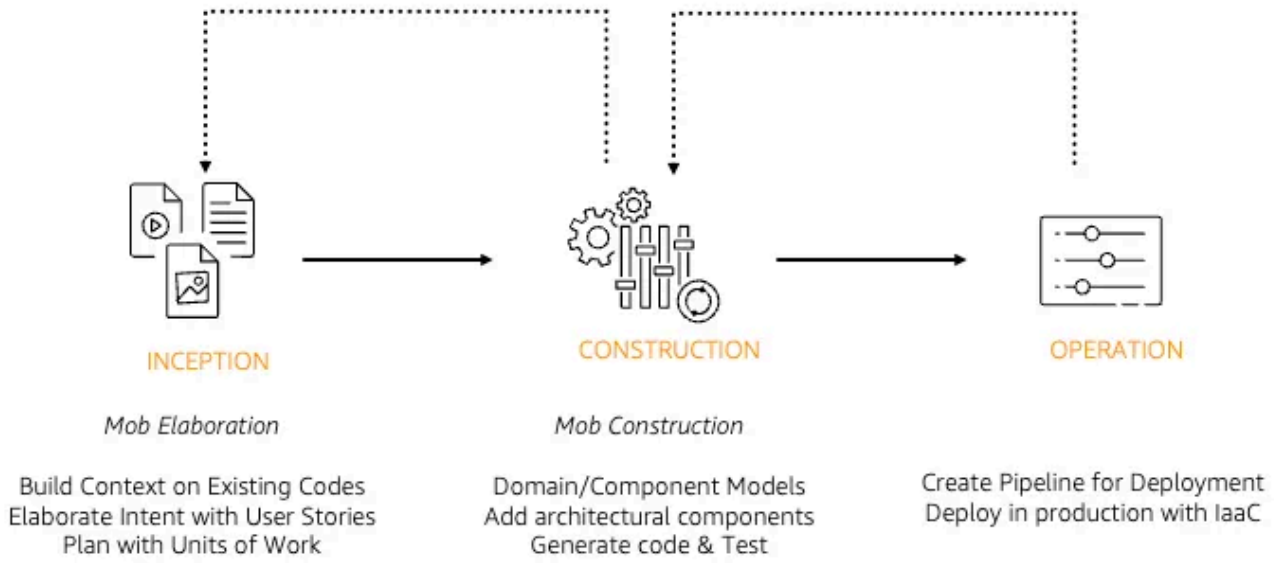


This pattern, where AI creates a plan, asks clarifying questions to seek context, and implements solutions only after receiving human validation, repeats rapidly for every SDLC activity, to provide a unified vision and approach for all development pathways.

With this mental model at its core, the software development in AI-DLC takes place in three straightforward phases:

- In the **Inception phase**, AI transforms business intent into detailed requirements, stories and units through "Mob Elaboration" – where the entire team actively validates AI's questions and proposals.
- In the **Construction phase**, using the validated context from the Inception phase, AI proposes a logical architecture, domain models, code solution and tests through "Mob Construction" – where the team provides clarification on technical decisions and architectural choices in real time.
- In the **Operations phase**, AI applies the accumulated context from previous phases to manage infrastructure as code and deployments, with team oversight.

Each phase provides richer context for the next, thus enabling AI to provide increasingly informed suggestions.



AI saves and maintains persistent context across all phases by storing plans, requirements, and design artifacts to your project repository, ensuring seamless continuation of work across multiple sessions.

AI-DLC introduces new terminology and rituals to reflect its AI-driven, highly collaborative approach. Traditional 'sprints' are replaced by 'bolts' – shorter, more intense work cycles measured in hours or days rather than weeks; Epics are replaced by Units of Work. This shift in terminology underscores the method's emphasis on speed and continuous delivery. Similarly, other familiar Agile terms are reimagined to align with the AI-centric workflow, creating a vocabulary that better represents the methodology's innovative approach to software development.

What are the benefits of this methodology?

- **Velocity**: The foremost benefit that AI-DLC offers is acceleration in development velocity, as AI rapidly generates and refines artifacts, such as requirements, stories, designs, code, and tests allowing product owners, architects, and developers to complete tasks in hours or days that previously took weeks.
- **Innovation**: Consequently, this acceleration and heavy lifting by AI, frees up significant time for innovation, enabling builders to explore creative solutions and push the boundaries of what's possible.

- **Quality**: With continuous clarification, teams build precisely what they have in mind, rather than an abstract AI interpretation of the intent. This results in products that are more closely aligned with business objectives. AI enhances quality by consistently applying organization-specific standards – your coding practices, design patterns, and security requirements – while generating comprehensive test suites. This end-to-end AI integration improves coherence and traceability from requirements to deployment.
- **Market Responsiveness**: The rapid development cycles of AI-DLC enable us to quickly respond to market demands and user feedback, and consequently faster adaptation to requirements.
- **Developer Experience**: AI-DLC transforms the developer experience by shifting focus from routine coding tasks to critical problem-solving. AI helps reduce cognitive load by handling repetitive tasks, while satisfaction is enhanced as developers gain deeper business context and witness how their work directly impacts business value.

How do I get started with this?

Begin your journey with AI-DLC, through three clear paths: Read the comprehensive AI-DLC white paper, explore how Amazon Q Developer rules and Kiro custom workflows can help you implement AI-DLC in your organization consistently or connect with your AWS account team to discuss how AI-DLC can be tailored to your organization's specific needs.

The future of software development is here. We are excited to help you leverage AI to not only build systems faster but also maintain fidelity and quality through critical human oversight and collaboration. Start your AI-DLC journey today and join the growing community of organizations transforming their development practices through AI-driven innovation.

Raja SP

Raja is a Principal Solutions Architect at AWS, where he leads Developer Transformation Programs. He has worked with more than 100 large customers, helping them design and deliver mission critical systems built on modern architectures, platform engineering practices, and Amazon inspired operating models. As generative AI reshapes the software development landscape, Raja and his team created the AI Driven Development Lifecycle (AI DLC) — an end to end, AI native methodology that reimagines how large teams collaboratively build production grade software in the AI era.